

Bridging the Observable Gap: SciML Architectures for Partially Observable Electromechanical Systems with Structural Model Error

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Abstract

Draft Abstract: This research investigates the "Closure Problem" in electromechanical energy harvesters, where high-dimensional analytical models ($n \approx 7$) must be reconciled with low-dimensional experimental observations ($m \approx 1$). Unlike traditional system identification, which assumes a correct model structure and tunes parameters, this work addresses **structural model error** ("missing physics") in the presence of **hidden dynamics**. We propose a cascade framework integrating Universal Differential Equations (UDEs), Port-Hamiltonian Neural ODEs, and Koopman Operator theory to reconstruct hidden states and learn missing physical terms while strictly enforcing thermodynamic constraints.

1 Introduction: The Reactivity and Closure Problem

1.1 The System Context

Electromechanical energy harvesters represent a class of "reactive" multi-domain dynamical systems. The system state $\mathbf{x}(t) \in \mathbb{R}^n$ evolves according to coupled mechanical and electrical laws, driven by inputs $\mathbf{u}(t)$ (e.g., vibration source). However, the physical realization of the device offers only a limited observation $\mathbf{y}(t) \in \mathbb{R}^m$, where $m \ll n$.

Typically, only voltage or tip displacement is measurable, leaving internal states (such as coupled modes, magnetic hysteresis states, or thermal gradients) hidden.

1.2 The Closure Problem

Standard engineering approaches assume the analytical model $f_{\text{analytical}}$ is structurally correct and only requires parameter estimation (θ). However, experimental evidence suggests the presence of **missing physics**—dynamics not captured by the derived constitutive equations.

The research problem is thus defined as a "Closure Problem": How do we learn a correction term $g_{\text{learn}}(\mathbf{x}, \mathbf{u})$ such that:

$$\dot{\mathbf{x}} = f_{\text{analytical}}(\mathbf{x}, \mathbf{u}; \theta) + g_{\text{learn}}(\mathbf{x}, \mathbf{u}) \quad (1)$$

subject to the constraint that $\mathbf{x}$ is only partially observable via $\mathbf{y} = h(\mathbf{x})$.

2 Baseline Methods: State Estimation & Control Theory

Note: This section covers the advisor’s recommendations as the ”Classical Baseline.”

2.1 Kalman Filtering Variants (EKF/UKF)

State estimation techniques like the Extended Kalman Filter (EKF) utilize a predictor-corrector architecture to estimate hidden states $\hat{\mathbf{x}}$.

- **Strength:** Well-suited for noise reduction when the model structure $f_{analytical}$ is accurate.
- **Limitation:** In the presence of structural error (missing physics), the Kalman Gain often ”smears” the error into the state estimate, leading to bias rather than identifying the missing dynamic term.

2.2 Nonlinear Model Predictive Control (NMPC)

[TODO: Discuss NMPC utility for input-output mapping, but note its computational cost and reliance on model accuracy.]

3 SciML Frontiers: Investigated Avenues

3.1 Universal Differential Equations (UDEs) & Adjoint Optimization

To address the ”missing physics,” we investigate UDEs where a Neural Network (NN) approximates the unknown term g_{learn} .

$$\dot{\mathbf{x}} = f_{physics} + \text{NN}(\mathbf{x}) \quad (2)$$

The Partial Observability Challenge: Standard UDE training requires full state data. We propose utilizing **Adjoint Sensitivity Analysis** to backpropagate the error from the output trajectory $\mathbf{y}(t)$ through the ODE solver to update the NN weights, effectively ”inferring” the influence of hidden states on the output.

3.2 Latent Dynamics & Koopman Operator (HAVOK)

Since the true dimension of the ”active” physics is unknown, we employ Hankel Alternative View of Koopman (HAVOK) analysis. By constructing Time-Delay Embeddings from the single output $\mathbf{y}(t)$, we can reconstruct a diffeomorphic shadow manifold of the full system.

- This validates if the ”7-state” model is necessary or if the dynamics live on a lower-dimensional attractor.
- Allows for linear embeddings of nonlinear phenomena.

4 Proposed Novelty: Energy-Constrained and Joint Architectures

Note: This section targets the ”high value / not yet investigated” areas.

4.1 Port-Hamiltonian Neural ODEs (PH-NODE)

A critical flaw in standard Black-Box Neural ODEs is the violation of conservation laws (e.g., generating free energy). For an energy harvester, this is physically inadmissible.

We propose a **Port-Hamiltonian framework** where the Neural Network learns the Hamiltonian (Energy function) $\mathcal{H}(\mathbf{x})$ and the Damping matrix $R(\mathbf{x})$, rather than the vector field directly:

$$\dot{\mathbf{x}} = [J(\mathbf{x}) - R(\mathbf{x})]\nabla\mathcal{H}(\mathbf{x}) + G(\mathbf{x})\mathbf{u} \quad (3)$$

This guarantees stability and passivity by construction, effectively constraining the "search space" for the missing physics to only physically realizable models.

4.2 Multi-Fidelity Transfer Learning

To overcome the lack of hidden state data, we propose a **Sim-to-Real Transfer** approach:

1. **Pre-training:** Train the UDE structure on high-fidelity FEM simulation data (where all $\mathbf{x}$ are known).
2. **Transfer:** Freeze the learned structure of the hidden couplings.
3. **Fine-tuning:** Use experimental output data $\mathbf{y}$ to tune only the coefficients of the missing physics term via Adjoint Optimization.

5 Methodology Plan

1. **Data Collection:** Electromechanical harvester response under stochastic and harmonic excitation.
2. **Topological Analysis:** Apply HAVOK/Takens Embedding to determine intrinsic dimensionality.
3. **Baseline Comparison:** Implement EKF using the analytical 7-state model. Quantify residuals.
4. **SciML Implementation:** Develop the Port-Hamiltonian UDE framework in Julia (SciML ecosystem).